La Scuola d'Italia / Dr. Huson / IB Math 12
20 October 2026

Nickname:

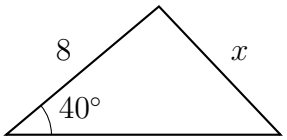
2.2 Classwork: The Cosine Rule

In triangle ABC, side a is opposite angle A, side b is opposite angle B, and so on.

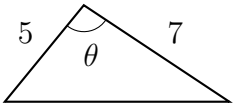
$$c^2 = a^2 + b^2 - 2ab\cos C \qquad \cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

Give answers correct to three significant figures, unless they are exact.

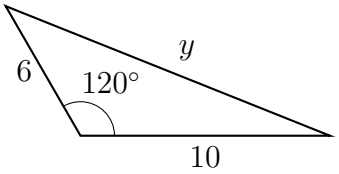
1. The diagrams are not to scale.



(a)



(b), (c)



(d)

(a) Find the value of x.

(b) Find the size of the largest angle, θ, in the second triangle.

(c) State whether the second triangle is acute, right-angled or obtuse.

(d) Find the exact value of y.

2. Two hikers leave a campsite at the same time and walk along straight paths. The angle between their paths is 35°. When the first hiker has walked 6 km, the second hiker has walked 9 km.

   (a) Find the distance between the two hikers at this time.

The quadrilateral $ABCD$ has $AB = 5\,\text{cm}$, $BC = 8\,\text{cm}$, $CD = 6\,\text{cm}$, $AD = 4\,\text{cm}$ and $\angle ABC = 60°$.

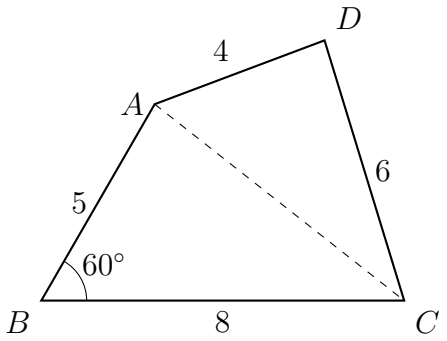


diagram not to scale

   (b) Find the length of the diagonal AC.

   (c) Find the size of angle ADC.